

Mathematical Analysis II

Week 8 Homework (April 23)

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2026 年 4 月 26 日

Section 10.2

1. 计算下列积分.

$$(1) \int_0^R dx \int_0^{\sqrt{R^2-x^2}} \ln(1+x^2+y^2) dy;$$

解. 令 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$, 所以 $dx dy = r dr d\theta$

$$\text{所以 } \int_0^R dx \int_0^{\sqrt{R^2-x^2}} \ln(1+x^2+y^2) dy = \int_0^{\frac{\pi}{2}} d\theta \int_0^R r \ln(1+r^2) dr$$

其中

$$\begin{aligned} \int_0^R r \ln(1+r^2) dr &= \int_0^R \frac{1}{2} \ln(1+r^2) d(1+r^2) \\ &\stackrel{u=1+r^2}{=} \frac{1}{2} (u \ln u - u) \Big|_0^R \\ &= \frac{1}{2} ((1+R^2) \ln(1+R^2) - 1 - R^2) \Big|_0^R \\ &= \frac{1}{2} ((1+R^2) \ln(1+R^2) - R^2) \end{aligned}$$

所以

$$\begin{aligned} \int_0^R dx \int_0^{\sqrt{R^2-x^2}} \ln(1+x^2+y^2) dy &= \int_0^{\frac{\pi}{2}} \frac{1}{2} ((1+R^2) \ln(1+R^2) - R^2) d\theta \\ &= \frac{\pi}{4} ((1+R^2) \ln(1+R^2) - R^2) \end{aligned}$$

$$(3) \int_0^\pi \int_0^\pi \cos(x+y) dx dy;$$

解.

$$\begin{aligned}\text{原式} &= \int_0^\pi \sin(x+y)|_0^\pi dy \\ &= -2 \int_0^\pi \sin y dy \\ &= 2 (\cos y)|_0^\pi \\ &= -4\end{aligned}$$

$$(5) \int_0^{\frac{R}{\sqrt{1+R^2}}} dx \int_0^{Rx} \left(1 + \frac{y^2}{x^2}\right) dy + \int_{\frac{R}{\sqrt{1+R^2}}}^R dx \int_0^{\sqrt{R^2-x^2}} \left(1 + \frac{y^2}{x^2}\right) dy;$$

解. 即求 $y = Rx$, $x^2 + y^2 = R^2$, $y = 0$ 围成的扇形面积

$$\text{令} \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

θ 范围: $0 \leq \theta \leq \arctan R$

$$\text{所以} \begin{cases} 1 + \frac{y^2}{x^2} = 1 + \cot^2 \theta = \sec^2 \theta \\ dx dy = r dr d\theta \end{cases}$$

所以

$$\begin{aligned}\text{原式} &= \int_0^{\arctan R} d\theta \int_0^R r \sec^2 \theta dr \\ &= \int_0^{\arctan R} \frac{r^2}{2} \sec^2 \theta \Big|_0^R d\theta \\ &= \frac{R^2}{2} \int_0^{\arctan R} \sec^2 \theta d\theta \\ &= \frac{R^2}{2} (\tan \theta) \Big|_0^{\arctan R} \\ &= \frac{R^3}{2}\end{aligned}$$

2. 计算下面二重积分.

$$(1) \iint_D \sqrt{x^2 + y^2} dx dy, D: x^2 + y^2 \leq x + y;$$

$$\text{解. 令} \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}, \text{ 所以 } dx dy = r dr d\theta, 0 \leq r \leq x + y = (\cos \theta + \sin \theta) = \sqrt{2} \sin(\theta + \frac{\pi}{4})$$

$$\text{所以 } 0 \leq \theta + \frac{\pi}{4} \leq \pi, \text{ 所以 } -\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$$

所以

$$\begin{aligned}
\text{原式} &= \iint_D r^2 \, dr \, d\theta \\
&= \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} d\theta \int_0^{\sqrt{2}\sin(\theta+\frac{\pi}{4})} r^2 \, dr \\
&= \frac{2\sqrt{2}}{3} \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \sin^3(\theta + \frac{\pi}{4}) \, d\theta
\end{aligned}$$

令 $\alpha = \theta + \frac{\pi}{4}$, 所以

$$\begin{aligned}
\text{原式} &= \frac{2\sqrt{2}}{3} \int_0^\pi \sin^3 \alpha \, d\alpha \\
&= \frac{4\sqrt{2}}{3} \int_0^{\frac{\pi}{2}} \sin^3 \alpha \, d\alpha \\
&= \frac{8\sqrt{2}}{9}
\end{aligned}$$

(3) $\iint_D (x^2 + y^2) \, dx \, dy$, D : 由 $xy = 1$, $xy = 2$, $y = x$, $y = 2x$ 围成的第一象限部分;

解. 令 $\begin{cases} u = xy \\ v = \frac{y}{x} \end{cases}$

$$\text{所以 } \begin{cases} 1 \leq u \leq 2 \\ 1 \leq v \leq 2 \end{cases}, \begin{cases} x^2 = \frac{u}{v} \\ y^2 = uv \end{cases}, \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} = \frac{y}{x} + \frac{y}{x} = \frac{2y}{x} = 2v$$

$$\text{所以 } dx \, dy = \frac{1}{2v} \, du \, dv$$

所以

$$\begin{aligned}
\text{原式} &= \int_1^2 dv \int_1^2 \frac{u}{2v^2} (1 + v^2) \, du \\
&= \frac{3}{2} \int_1^2 \frac{1}{2v^2} + \frac{1}{2} \, dv \\
&= \frac{3}{4} \left(v - \frac{1}{v} \right) \Big|_1^2 \\
&= \frac{9}{8}
\end{aligned}$$

(5) $\iint_D xy \, dx \, dy$, D : $xy = a$, $xy = b$, $y^2 = cx$, $y^2 = dx$ 围成的第一象限部分 ($0 < a < b$, $0 < c < d$);

解. 令 $\begin{cases} u = xy \\ v = \frac{y^2}{x} \end{cases}$

所以 $\begin{cases} a \leq u \leq b \\ c \leq v \leq d \end{cases}$, $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} y & x \\ -\frac{y^2}{x^2} & \frac{2y}{x} \end{vmatrix} = \frac{2y^2}{x} + \frac{y^2}{x} = \frac{3y^2}{x} = 3v$

所以 $dx dy = \frac{1}{3v} du dv$

所以

$$\begin{aligned} \text{原式} &= \int_c^d dv \int_a^b \frac{u}{3v} du \\ &= \frac{b^2 - a^2}{6} \int_c^d \frac{1}{v} dv \\ &= \frac{b^2 - a^2}{6} \ln \frac{d}{c} \end{aligned}$$

(7) $\iint_D \frac{x^2 - y^2}{\sqrt{x+y+3}} dx dy$, $D: |x| + |y| \leq 1$;

解. 令 $\begin{cases} u = x + y \\ v = x - y \end{cases}$

所以 $\begin{cases} -1 \leq u \leq 1 \\ -1 \leq v \leq 1 \end{cases}$, $\begin{cases} x^2 - y^2 = uv \\ \sqrt{x+y+3} = \sqrt{u+3} \end{cases}$, $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$

所以 $dx dy = \frac{1}{2} du dv$

所以

$$\begin{aligned} \text{原式} &= \frac{1}{2} \int_{-1}^1 dv \int_{-1}^1 \frac{uv}{\sqrt{u+3}} du \\ &= \frac{1}{2} \int_{-1}^1 dv \int_{-1}^1 v \left(\sqrt{u+3} - \frac{3}{\sqrt{u+3}} \right) du \\ &= \frac{1}{2} \int_{-1}^1 v \left[\frac{2(u+3)^{\frac{3}{2}}}{3} - 6\sqrt{u+3} \right]_{-1}^1 dv \\ &= \frac{1}{2} \left(\frac{16}{3} - 12 - \frac{4\sqrt{2}}{3} + 6\sqrt{2} \right) \cdot 0 \\ &= 0 \end{aligned}$$

(9) $\iint_D |xy| dx dy$, $D = \{(x, y) : x^2 + y^2 \leq a^2\}$;

解. 令 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$, 所以 $dx dy = r dr d\theta$

原式为第一象限扇形面积的 4 倍, 所以

$$\begin{aligned}
 \text{原式} &= 4 \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta \, d\theta \cdot \int_0^a r^2 \, dr \\
 &= 4 \cdot \left. \frac{\sin^2 \theta}{2} \right|_0^{\frac{\pi}{2}} \cdot \frac{a^4}{4} \\
 &= \frac{a^4}{2}
 \end{aligned}$$

3. 求下列曲线所围成的平面区域的面积:

(1) $x^2 + 2y^2 = 3$ 和 $xy = 1$ (不含原点部分);

解. 联立求解: $x^2 + \frac{2}{x^2} = 3$, 解得第一象限内交点 $(1, 1), (\sqrt{2}, \frac{\sqrt{2}}{2})$

因为围成不含原点的平面区域为关于原点中心对称的两部分, 只需计算第一象限内的部分面积并乘 2.

在第一象限内:
$$\begin{cases} y = \sqrt{\frac{3-x^2}{2}} \\ y = \frac{1}{x} \end{cases}$$

所以

$$\begin{aligned}
 \text{第一象限面积} &= \int_1^{\sqrt{2}} \sqrt{\frac{3-x^2}{2}} - \frac{1}{x} \, dx \\
 &= \frac{\sqrt{2}x}{4} \sqrt{3-x^2} + \frac{3\sqrt{2}}{4} \arcsin \frac{x}{\sqrt{3}} - \ln x \Big|_1^{\sqrt{2}}
 \end{aligned}$$

所以

$$\begin{aligned}
 \text{面积} &= \frac{\sqrt{2}x}{2} \sqrt{3-x^2} + \frac{3\sqrt{2}}{2} \arcsin \frac{x}{\sqrt{3}} - \ln x^2 \Big|_1^{\sqrt{2}} \\
 &= (1 + \frac{3\sqrt{2}}{2} \arcsin \sqrt{\frac{2}{3}} - \ln 2) - (1 + \frac{3\sqrt{2}}{2} \arcsin \sqrt{\frac{1}{3}}) \\
 &= \frac{3\sqrt{2}}{2} (\arcsin \sqrt{\frac{2}{3}} - \arcsin \sqrt{\frac{1}{3}}) - \ln 2
 \end{aligned}$$

(2) $(x-y)^2 + x^2 = a^2$ ($a > 0$);

解. 令 $\begin{cases} u = x - y \\ v = x \end{cases}$, 所以 $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = 1$, 所以 $dx dy = du dv$

令 $\begin{cases} u = r \cos \theta \\ v = r \sin \theta \end{cases}$, 所以 $du dv = r dr d\theta$

所以即求 $\int_0^{2\pi} d\theta \int_0^a r dr = \pi a^2$

(3) 由直线 $x + y = a$, $x + y = b$, $y = kx$ 和 $y = mx$ ($0 < a < b$, $0 < k < m$) 围成的平面区域.

解. 令 $\begin{cases} u = x + y \\ v = \frac{y}{x} \end{cases}$, 所以 $\begin{cases} a \leq u \leq b \\ k \leq v \leq m \end{cases}$, $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 1 \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} = \frac{x+y}{x^2} = \frac{(1+v)^2}{u}$
 所以 $dx dy = \frac{u}{(1+v)^2} du dv$
 所以面积即

$$\begin{aligned} \int_k^m dv \int_a^b \frac{u}{(1+v)^2} du &= \frac{b^2 - a^2}{2} \int_k^m \frac{1}{(1+v)^2} dv \\ &= \frac{b^2 - a^2}{2} \left(\frac{1}{1+k} - \frac{1}{1+m} \right) \\ &= \frac{(a+b)(a-b)(k-m)}{2(1+k)(1+m)} \end{aligned}$$

4. 证明 $\iint_{x^2+y^2 \leq 1} e^{x^2+y^2} dx dy \leq \left[\int_{-\frac{\sqrt{\pi}}{2}}^{\frac{\sqrt{\pi}}{2}} e^{x^2} dx \right]^2$.

证. LHS: 在单位圆内积分, 记圆为 C

RHS: 在以原点为中心, 边长为 $\sqrt{\pi}$ 的正方形内积分, 记正方形为 S

$$\text{RHS} = \iint_{\substack{-\frac{\sqrt{\pi}}{2} \leq x \leq \frac{\sqrt{\pi}}{2} \\ -\frac{\sqrt{\pi}}{2} \leq y \leq \frac{\sqrt{\pi}}{2}}} e^{x^2+y^2} dx dy$$

$$\text{所以 } |C| = |S| = \pi$$

$$\text{记 } C \cap S = D, \text{ 所以 } \begin{cases} C = C \setminus D + D = C \setminus S + D \\ S = S \setminus D + D = S \setminus C + D \end{cases}$$

$$\text{所以 LHS} = \iint_{C \setminus S} e^{x^2+y^2} dx dy + \iint_D e^{x^2+y^2} dx dy$$

$$\text{RHS} = \iint_{S \setminus C} e^{x^2+y^2} dx dy + \iint_D e^{x^2+y^2} dx dy$$

$$\text{所以即证 } \iint_{C \setminus S} e^{x^2+y^2} dx dy \leq \iint_{S \setminus C} e^{x^2+y^2} dx dy$$

$$\text{在 } C \setminus S \text{ 内, } x^2 + y^2 \leq 1, \text{ 所以 } \iint_{C \setminus S} e^{x^2+y^2} dx dy \leq e|C \setminus S|$$

$$\text{在 } S \setminus C \text{ 内, } x^2 + y^2 \geq 1, \text{ 所以 } \iint_{S \setminus C} e^{x^2+y^2} dx dy \geq e|S \setminus C|$$

又因为 $|C \setminus S| = |S \setminus C|$, 所以

$$\begin{aligned} \iint_{C \setminus S} e^{x^2+y^2} dx dy &\leq e|C \setminus S| \\ &= e|S \setminus C| \\ &\leq \iint_{S \setminus C} e^{x^2+y^2} dx dy \end{aligned}$$

所以 $\text{LHS} \leq \text{RHS}$

Q.E.D.

5. 设 $f(x)$ 在 $[0, 1]$ 上连续, 证明 $\int_0^1 e^{f(x)} dx \cdot \int_0^1 e^{-f(y)} dy \geq 1$.

证. 令 $\begin{cases} u(x) = e^{\frac{f(x)}{2}} \\ v(y) = e^{-\frac{f(y)}{2}} \end{cases}$
所以

$$\begin{aligned} \int_0^1 e^{f(x)} dx \cdot \int_0^1 e^{-f(y)} dy &= \left(\int_0^1 u^2(x) dx \right) \cdot \left(\int_0^1 v^2(y) dy \right) \\ &= \left(\int_0^1 u^2(x) dx \right) \cdot \left(\int_0^1 v^2(x) dx \right) \\ &\geq \left(\int_0^1 u(x)v(x) dx \right)^2 \\ &= \left(\int_0^1 e^{\frac{f(x)}{2} - \frac{f(x)}{2}} dx \right)^2 \\ &= \left(\int_0^1 e^0 dx \right)^2 \\ &= \left(\int_0^1 1 dx \right)^2 \\ &= 1 \end{aligned}$$

Q.E.D.

6. 设 $f(x)$ 为连续的奇函数, 证明 $\iint_{|x|+|y|\leq 1} e^{f(x+y)} dx dy \geq 2$.

证. 令 $\begin{cases} u = x + y \\ v = x - y \end{cases}$, 所以 $\begin{cases} -1 \leq u \leq 1 \\ -1 \leq v \leq 1 \end{cases}$, $dx dy = \frac{1}{2} du dv$
所以

$$\begin{aligned} \text{LHS} &= \frac{1}{2} \int_{-1}^1 du \int_{-1}^1 e^{f(u)} dv \\ &= \int_{-1}^1 e^{f(u)} du \end{aligned}$$

因为 $f(x)$ 是连续的奇函数, 所以

$$\begin{aligned}
\text{LHS} &= \int_{-1}^1 e^{f(u)} du \\
&= \int_{-1}^0 e^{f(u)} du + \int_0^1 e^{f(u)} du \\
&= \int_0^1 e^{-f(u)} du + \int_0^1 e^{f(u)} du \\
&= \int_0^1 e^{-f(u)} + e^{f(u)} du \\
&\geq \int_0^1 2\sqrt{e^{-f(u)} \cdot e^{f(u)}} du \\
&= \int_0^1 2\sqrt{e^0} du \\
&= \int_0^1 2 du \\
&= 2
\end{aligned}$$

Q.E.D.

7. 设 $f(t)$ 为连续函数, 求证

$$\iint_D f(x-y) dx dy = \int_{-A}^A f(t) (A - |t|) dt$$

其中 D 为: $|x| \leq \frac{A}{2}, |y| \leq \frac{A}{2}, A > 0$ 为常数.

证. 令 $\begin{cases} u = x + y \\ v = x - y \end{cases}$, 所以 $|u| + |v| \leq A, dx dy = \frac{1}{2} du dv$
 所以令 $-A \leq v \leq A, -(A - |v|) \leq u \leq A - |v|$
 所以

$$\begin{aligned}
\text{LHS} &= \frac{1}{2} \int_{-A}^A dv \int_{-(A-|v|)}^{A-|v|} f(v) du \\
&= \frac{1}{2} \int_{-A}^A 2f(v)(A - |v|) dv \\
&= \int_{-A}^A f(v)(A - |v|) dv
\end{aligned}$$

将 v 替换成 t , 即得证

Q.E.D.